

STRESS-TESTING B-FAIS: FORECAST-AWARE BLOCK-WISE IMPUTER SELECTION FOR TIME-SERIES FOUNDATION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Pretrained time-series forecasters require finite numerical contexts, which makes imputation part of the forecasting pipeline. Block-wise Forecast-Aware Imputer Selection (B-FAIS) assigns an imputer to each contiguous missing run using downstream forecast loss, pseudo-missing evidence, a sparse block graph, and validity-gated search. This paper tests whether the added mechanisms provide measurable value. We pre-register six comparisons and evaluate disjoint training and confirmation masks over 26 dataset versions from 16 non-ETT families, with 2,340 paired episodes per forecaster. Five router seeds, 5,000-replicate hierarchical intervals, and Holm correction are retained. Forecast supervision versus reconstruction supervision gives family-macro MASE differences of -0.087 $[-0.284, 0.053]$ for Chronos-2 and $+0.052$ $[-0.078, 0.229]$ for TimesFM 2.5. Independent block routing versus sequence routing gives -0.029 $[-0.116, 0.040]$ and -0.038 $[-0.120, 0.015]$; adding graph-based coordination changes MASE by only -0.001 and -0.003 , with both intervals crossing zero. Leave-one-family-out and held-out-Sundial comparisons also fail the registered superiority rule. Two fixed tests are unavailable because pre-specified completeness conditions fail. On 72 naturally incomplete Beijing air-quality episodes, B-FAIS is worse than the best complete fixed candidate by 0.012 – 0.049 MASE across the three forecasters. The study does not validate forecast supervision, block granularity, structured coordination, or transfer as general contributions. It identifies a simpler sequence-level selector as the appropriate reference point for further work.

1 INTRODUCTION

Time-series foundation models (TSFMs) make zero-shot forecasting practical across datasets and sampling frequencies. TimesFM uses decoder-only patch prediction (Das et al., 2024), Chronos maps scaled observations to tokens for probabilistic forecasting (Ansari et al., 2024), and Chronos-2 extends pretrained forecasting to grouped multivariate inputs (Ansari et al., 2025). Their parameters can remain frozen at deployment, yet the numerical context still determines the forecast. Missing intervals alter local continuity, seasonal evidence, cross-variable relations, and the range of values received by the forecaster.

No single imputation rule is uniformly suited to these conditions. Interpolation assumes local smoothness, seasonal copying assumes repetition, multivariate completion assumes shared structure, and specialized neural imputers learn different temporal priors (van Buuren & Groothuis-Oudshoorn, 2011; Stekhoven & Bühlmann, 2012; Mazumder et al., 2010; Cao et al., 2018; Tashiro et al., 2021; Du et al., 2023; Nie et al., 2024). Reconstruction accuracy can also disagree with downstream forecast accuracy (Wang et al., 2024). These observations motivate algorithm selection conditioned on the corrupted context and the frozen forecaster (Rice, 1976; Mısırlı & Sebag, 2017; Zhao et al., 2021; Hazimeh et al., 2021; Zhou et al., 2020).

B-FAIS was designed for this setting. It represents each maximal contiguous missing run as a block, predicts candidate risks from block and episode features, measures candidates on pseudo-hidden observed values, and solves a validity-constrained assignment. A sparse graph and pairwise risk model permit different imputers to coordinate across nearby or correlated blocks. The design is

plausible, though each added mechanism creates a separate empirical claim. Forecast labels must improve on reconstruction labels. Block decisions must improve on one sequence decision. Graph-based coordination must improve on independent block choices. Training labels must also transfer across data families and, for a model-agnostic claim, across forecasters.

The earlier evaluation of the complete system combined these mechanisms and compared it with sequence selectors in a rolling-origin setting (Tashman, 2000). That design established system-level behavior within observed datasets, while causal attribution among the mechanisms remained unresolved. We therefore conduct an R2 evaluation that fixes the questions, splits, seeds, statistics, and interpretation rules before inspecting confirmation results. The comparison ladder changes one factor at a time: SEQ-RECON, SEQ-FORECAST, IND-BLOCK-FORECAST, and STRUCT-BLOCK-FORECAST. ETT is restricted to development. Confirmation uses disjoint masks on 16 non-ETT families, and leave-one-family-out (LOFO) training evaluates unseen-family behavior. Sundial (Liu et al., 2025) is held out from teacher-label generation and hyperparameter selection.

The evidence is consistently cautious. None of the six registered tests satisfies the joint superiority rule after Holm correction. Four tests are complete and yield intervals that cross zero; two are incomplete because pre-specified availability rules prevent the intended comparison. A naturally incomplete air-quality dataset provides an additional negative check. These outcomes change the paper’s contribution. The main contribution is a controlled empirical account of which parts of forecast-aware block routing remain unsupported under a stronger protocol, together with machine-readable evidence for unavailable tests and tail failures. The resulting recommendation is to treat sequence-level selection as the primary reference and to require new evidence before restoring stronger mechanism or transfer claims.

2 RELATED WORK

Pretrained forecasting. TimesFM predicts output patches from input patches and supports inference without target-dataset fitting (Das et al., 2024). Chronos tokenizes scaled values and samples future paths from a pretrained model (Ansari et al., 2024). Chronos-2 uses grouped attention for multivariate and covariate-informed inputs (Ansari et al., 2025). Sundial learns a continuous generative forecaster using a flow-matching objective (Liu et al., 2025). Our evaluation keeps these checkpoints frozen. TimesFM 2.5 and Sundial are invoked independently per target; Chronos-2 receives a joint multivariate context. These interface differences make forecaster-specific imputation behavior plausible and motivate separate reporting.

Time-series imputation. Classical candidates include chained equations, random-forest completion, low-rank completion, and temporally regularized factorization (van Buuren & Groothuis-Oudshoorn, 2011; Stekhoven & Bühlmann, 2012; Mazumder et al., 2010; Yu et al., 2016). Neural candidates include bidirectional recurrence in BRITS (Cao et al., 2018), latent Gaussian-process structure in GP-VAE (Fortuin et al., 2020), conditional diffusion in CSDI (Tashiro et al., 2021), self-attention in SAITS (Du et al., 2023), and spatiotemporal attention in ImputeFormer (Nie et al., 2024). General time-series models such as TimeMixer++ and TOTEM have also been evaluated for imputation (Wang et al., 2025; Talukder et al., 2024). HELIX explicitly models variable identity and cross-dimensional information (Zhang et al., 2026). B-FAIS leaves each candidate objective unchanged and evaluates its completed context through a frozen forecaster.

Algorithm selection and downstream utility. Algorithm selection maps instance descriptors to a candidate under a task-specific performance measure (Rice, 1976). ALORS learns task–algorithm factors (Mısır & Sebag, 2017); MetaOD predicts a latent task representation (Zhao et al., 2021); DSelect- k constructs a sparse differentiable gate (Hazimeh et al., 2021); and NeuralUCB combines predicted reward with uncertainty (Zhou et al., 2020). HybridLSTM recommends imputers for fixed univariate windows (Almeida et al., 2025). Task-oriented imputation directly evaluates downstream forecasting consequences (Wang et al., 2024). Our controlled ladder connects these lines by separating the training target from the decision granularity and the assignment structure.

3 PROBLEM SETUP

Let $X \in \mathbb{R}^{T \times D}$ be a complete trajectory and $M \in \{0, 1\}^{T \times D}$ an observation mask. At forecast origin t , a corrupted context and clean future are

$$C_t = (M \odot X)[t - L : t], \quad Y_t = X[t : t + H, \mathcal{K}], \quad (1)$$

where missing entries of C_t are represented as NaN and $\mathcal{K}$ is the target set. Candidate $a \in \mathcal{A}$ maps C_t to a completion $I_a(C_t)$. A separate NATIVE-VALID mask records cells produced by the candidate before numerical repair. The frozen forecaster f maps a finite completion to $\hat{Y}$.

For target k , the MASE scale s_k is computed from the historical fit prefix with its registered seasonal lag q_k , or lag one when seasonal history is insufficient (Hyndman & Koehler, 2006). The episode loss is

$$\mathcal{L}_f(Z, Y) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{H^{-1} \sum_{h=1}^H |Y_{h,k} - f(Z)_{h,k}|}{s_k}. \quad (2)$$

Every maximal contiguous missing run in one variable is a block $b = (d, s_b, e_b)$. A block assignment $\mathbf{a} = (a_b)_{b \in \mathcal{B}}$ assembles $Z(\mathbf{a})$ by copying candidate a_b on block b and restoring observed cells. Deployment chooses $\mathbf{a}$ without access to Y_t .

4 B-FAIS AND THE CONTROLLED VARIANTS

Figure 1 separates the historical supervision path from deployment and aligns the deployed components with the registered controls. At historical origins, the teacher evaluates complete-candidate forecasts against observed futures and derives second-order pair effects. At deployment, the router extracts maximal missing blocks, produces a safety-augmented shortlist, refines candidate risks using the actual context and pseudo-missing runs, and then applies a sparse relation graph, a native-validity gate, and deterministic beam search. The selected block outputs are assembled into the finite context passed to the frozen forecaster.

4.1 FORECAST-AWARE LABELS AND BLOCK SCORES

At historical training origins, the teacher evaluates the complete context generated by candidate a :

$$\ell_{e,a}^{\text{full}} = \mathcal{L}_f(I_a(C_e), Y_e). \quad (3)$$

The R2 target audit selected this complete-candidate forecast loss on the ETT development family over a single-block counterfactual loss and a coordination-adjusted target. For each sampled labelled block, $\ell_{e,a}^{\text{full}}$ is attached to the candidate row. Block geometry, missingness, candidate descriptors, and available context features allow a LambdaMART ranker (Burges, 2010), implemented with LightGBM (Ke et al., 2017), to learn a block-conditioned score, although the supervised target remains a complete-context outcome. Dataset identity, family identity, missingness-generator identity, and the generator’s target rate are removed from the deployment feature set.

The prior scorer creates a shortlist of at most six candidates, with LOCF and linear interpolation retained as safety candidates. Shortlisted candidates are run on the actual context and on eight pseudo-missing blocks sampled from observed history. The refined LambdaMART scorer receives native coverage, reconstruction error, bias, output summaries, block descriptors, and forecaster descriptors. The final R2 core uses this learned refined score directly. Development-only mixtures with a global candidate prior and a raw pseudo-error proxy were worse on ETT and are excluded from confirmation. Candidate runtime affects shortlist construction; the assignment objective has zero runtime weight.

4.2 INDEPENDENT AND STRUCTURED BLOCK ASSIGNMENT

For structured routing, a sparse graph connects same-variable neighbors, temporally overlapping blocks, and nearby blocks on correlated variables. A Huber regressor estimates pair interactions from sampled second-order counterfactuals. The structured energy is

$$E(\mathbf{a}) = \sum_{b \in \mathcal{B}} R(b, a_b) + \beta \sum_{(b,c) \in \mathcal{E}} w_{bc} \hat{\Phi}_{bc}(a_b, a_c), \quad (4)$$

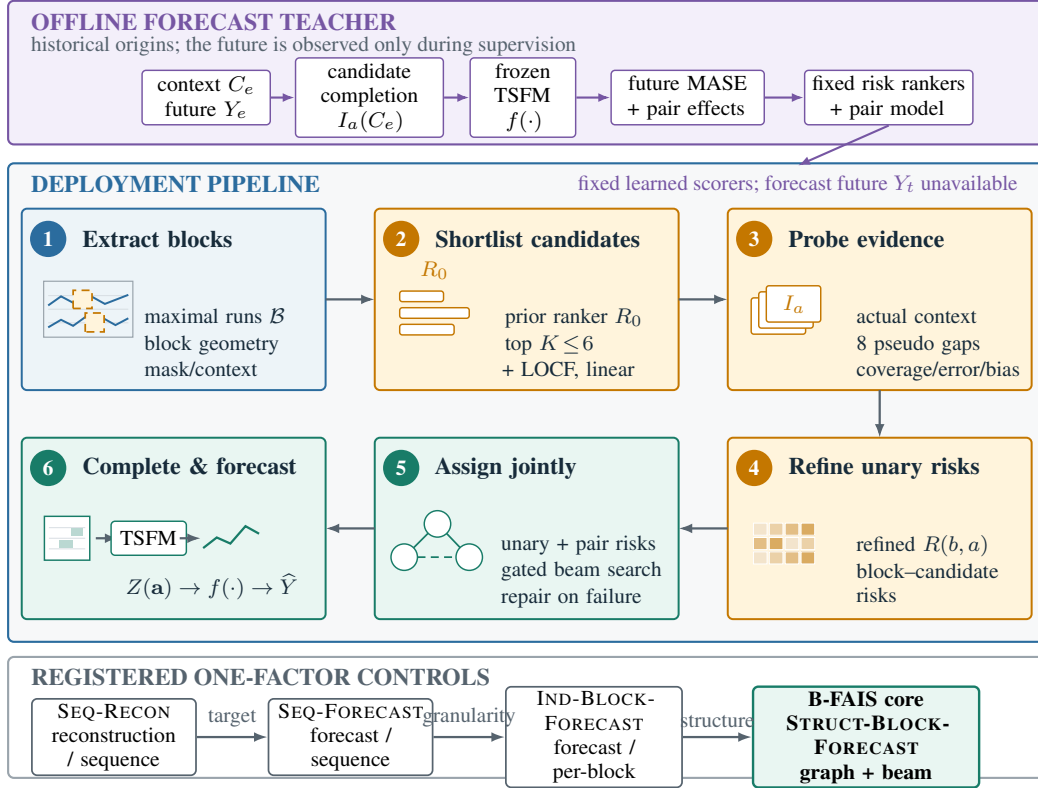


Figure 1: B-FAIS converts a corrupted multivariate context into a validity-gated block-wise completion before frozen-TSFM forecasting. At historical origins (top), complete-candidate forecast losses and second-order counterfactual effects train the unary rankers and pair model; at deployment (middle), six numbered steps extract blocks, shortlist candidates, collect pseudo-missing evidence, refine unary risks, coordinate the assignment, and assemble the completed context. The bottom ladder identifies the three one-factor changes used by the registered mechanism tests.

with $\beta = 1$. Deterministic beam search uses width 32. Independent block routing minimizes each unary score separately. Sequence routing selects one candidate using its mean score over the episode’s blocks. Candidate a can fill block b only when every cell in b is NATIVE-VALID. If no shortlisted candidate is valid, a separate repair state applies linear interpolation, LOCF, or the training median. Repair does not change candidate coverage.

4.3 ONE-FACTOR COMPARISON LADDER

The lower band of Figure 1 isolates the three registered mechanism changes. SEQ-RECON trains a sequence selector with masked-context reconstruction error; SEQ-FORECAST changes only the target to Equation 3; IND-BLOCK-FORECAST changes only the decision granularity; and STRUCT-BLOCK-FORECAST adds the relation graph, pair regressor, and coordinated search. The final B-FAIS core corresponds to STRUCT-BLOCK-FORECAST with the ETT-selected complete-candidate target and learned-only refined evidence.

5 EXPERIMENTAL PROTOCOL

Development and confirmation. The four ETT versions are restricted to target and evidence development. Rolling confirmation contains 26 dataset versions from 16 non-ETT families. Each dataset contributes 90 episodes per forecaster, giving 2,340 paired episodes for each adjacent comparison. Context length, horizon, and forecast stride are 96. Six missingness generators are crossed with rates $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ and confirmation mask seeds $\{3101, 3102, 3103\}$. Training uses seeds

Table 1: Adjacent mechanism tests on common-16 confirmation. Δ is first method minus reference MASE. p_w is the component Wilcoxon value. The joint value is the maximum component value for each test.

Test	Contrast	Forecaster	Δ	95% CI	p_w	$p_{\text{joint}}/p_{\text{Holm}}$
H1	Forecast — Recon	Chronos-2	−.087	[−.284, .053]	.083	.744 / 1.000
		TimesFM 2.5	+.052	[−.078, .229]	.744	
H2	Block — Sequence	Chronos-2	−.029	[−.116, .040]	.464	.464 / 1.000
		TimesFM 2.5	−.038	[−.120, .015]	.051	
H3	Structured — Independent	Chronos-2	−.001	[−.015, .009]	.464	.464 / 1.000
		TimesFM 2.5	−.003	[−.012, .003]	.348	

{1101, 1102, 1103} and ETT development uses {2101, 2102, 2103}. A full-sequence mask is created before rolling windows are sliced. Of the 2,340 confirmation episodes, 2,200 contain missing context cells; zero-block windows remain in the fixed all-window population. Earlier origins provide router labels and later origins provide evaluation. Details and the dataset inventory appear in Appendix A.

Forecasters, candidates, and external selectors. Chronos-2 and TimesFM 2.5 supply training labels and are evaluated separately. Each forecast uses 20 samples where the model interface is stochastic. A third checkpoint, `thuml/sundial-base-128m`, is held out from label generation and development. Its identity-free router removes the forecaster indicator. The registry contains 20 imputer entries, of which 19 execute under the rolling interface; TRMF is disabled because the installed wrapper is transductive. MetaOD, DSelect-1, NeuralUCB, ALORS, HybridLSTM, and random-valid selection retain their reconstruction-based training objective. They do not receive downstream forecast labels.

Transfer and natural missingness. LOFO retrains the router while withholding every labelled example, prior, and calibration value from the evaluated family. Its 17-family population adds the only eligible Job Claims origin and contains 2,430 episodes per forecaster. The held-out-Sundial test uses the common 16-family population. A separate test uses the UCI Beijing Multi-Site Air Quality dataset (Chen, 2017): 12 stations, 11 numerical variables, native missing entries, PM2.5 and PM10 targets, and 72 valid rolling episodes. No synthetic mask is added in this test.

Statistics and decision rule. The primary statistic is the equal-weight mean of family-level MASE. Effects are paired differences, with negative values favoring the method named first. Intervals use 5,000 hierarchical resamples over family, dataset, item, origin, and mask. Each component also receives a paired family-level Wilcoxon test. The six fixed tests are H1 forecast versus reconstruction, H2 block versus sequence, H3 structured versus independent, H4 core versus a development-selected external selector, H5 LOFO core versus LOFO sequence, and H6 Sundial core versus both an availability-qualified fixed imputer and identity-free sequence selection. A joint test uses the maximum required component p -value and is set to one when a required component is unavailable. Holm correction (Holm, 1979) covers all six joint tests. Superiority requires every required interval upper bound below zero and adjusted $p < 0.05$. Five router seeds, 4101–4105, are retained; seed 4101 is the registered primary analysis.

6 RESULTS

6.1 THE THREE PROPOSED MECHANISMS DO NOT PASS CONFIRMATION

Table 1 contains the six complete components for H1–H3. Forecast supervision has opposite aggregate signs: it reduces Chronos-2 MASE by 0.087 and increases TimesFM 2.5 MASE by 0.052. Both intervals include zero. Independent block routing has negative point estimates against sequence routing for both forecasters, with upper confidence limits of 0.040 and 0.015. Structured coordination changes family-macro MASE by less than 0.004. The component intervals cross zero, and every joint Holm value is one.

The absolute family-macro MASE of the structured core is 7.516 for Chronos-2 and 2.335 for TimesFM 2.5. The corresponding sequence-forecast values are 7.547 and 2.375. All five router seeds

Table 2: Transfer comparisons. The LOFO population contains 17 families; Sundial uses common-16. Negative values favor the structured core.

Test	Forecaster	Pairs	Δ	95% CI	p_w
H5: LOFO core – sequence	Chronos-2	2,430	-.018	[-.107, .082]	.378
	TimesFM 2.5	2,430	-.077	[-.192, .001]	.080
H6: held-out core – sequence	Sundial	2,340	-.016	[-.094, .059]	.940

Table 3: Natural-missing evaluation on 72 Beijing air-quality episodes. The best fixed column is restricted to candidates valid on all 72 episodes. Δ is B-FAIS minus that candidate.

Forecaster	B-FAIS	Best complete candidate	Best	Δ	Oracle
Chronos-2	1.012	TimeMixer++	1.000	+.012	.943
TimesFM 2.5	1.127	KNN multivariate	1.078	+.049	1.018
Sundial	.909	TimeMixer++	.880	+.029	.786

produce the same core family-macro values and the same saved assignments at the reported precision. This seed invariance removes one source of uncertainty in the fitted router, while the episode and family variation represented by the intervals remains.

6.2 UNSEEN-FAMILY AND UNSEEN-FORECASTER RESULTS REMAIN UNCERTAIN

LOFO point estimates favor the core, yet the intervals in Table 2 cross zero. TimesFM 2.5 comes closest, with an upper limit of +0.0012 and a component p -value of 0.080. The joint LOFO test is 0.378 before correction and one after correction. On held-out Sundial, identity-free structured routing differs from identity-free sequence routing by -0.016 with a wide interval and $p = 0.940$. These tests do not support unseen-family or cross-forecaster superiority.

H4 is unavailable. The pre-registered ETT selection rule required complete primary rows for every candidate external selector. HybridLSTM lacked 100 of 360 ETT episodes for each trained forecaster, so the guard stopped selection and no strongest external method was named. H6 is also incomplete because no fixed imputer met its pre-registered availability floor. Under the fixed rule, both tests receive joint and adjusted values of one. Descriptively, the core has lower common-16 family-macro MASE than each of the five pairable external selectors for both trained forecasters, with deltas ranging from -4.410 to -0.376 for Chronos-2 and from -3.615 to -0.464 for TimesFM 2.5. These descriptive results cannot replace the unavailable H4 test.

6.3 NATURAL MISSINGNESS DOES NOT RECOVER THE CLAIMED ADVANTAGE

Table 3 reports the UCI Beijing evaluation. B-FAIS covers all 72 episodes. Its mean MASE exceeds the best candidate that is complete on the same 72 episodes for every forecaster. The gaps are 0.012 for Chronos-2, 0.049 for TimesFM 2.5, and 0.029 for Sundial. The per-episode oracle remains lower, indicating unused selection headroom. This single-family analysis is descriptive and supplies no superiority test.

7 DISCUSSION

The controlled ladder changes the interpretation of B-FAIS. H1 does not establish forecast loss as a better training signal than reconstruction loss. The model-specific signs also warn against pooling TSFMs into one claim. H2 does not establish block granularity over one sequence decision. H3 provides the narrowest intervals and a near-zero effect, so the relation graph, pair regressor, and beam coordination lack evidence as performance contributions in this protocol. The pre-registered interpretation rules therefore remove these mechanisms from the central contribution claim.

The intervals do not establish equivalence. H1 and H2 still permit effects that may matter for some families, especially for TimesFM block routing. A suitable next experiment would target the families and missing geometries that drive this heterogeneity, then evaluate a fixed conditional rule on new

families. The current evidence supports no such rule. The simpler sequence selector is consequently the defensible operational reference, with independent block routing retained only as an experimental alternative.

The unavailable tests reveal a separate design lesson. A strict guard prevents favorable method selection after observing confirmation results, which preserves the registered decision. The same guard can make a multi-method comparison uninformative when one method has incomplete coverage. Future protocols should define a complete-case primary selector set and a separate coverage analysis before development evaluation. The fixed-imputer floor needs the same treatment.

The natural-missing test and prefix sensitivity further constrain deployment. When the historical fit prefix is forced to contain missing values, family-macro changes remain uncertain, while the worst episode degradation reaches 220.8 MASE for Chronos-2, 44.9 for TimesFM 2.5, and 39.9 for Sundial. Mean performance can therefore hide severe local failures. Candidate validity, tail metrics, and explicit fallback evaluation should remain first-class outputs even if a future selector improves the mean.

8 LIMITATIONS

The confirmation collection uses controlled synthetic masks. It spans six generators and five rates, yet those generators cannot represent every sensor, logging, or operational failure process. The natural-missing study contains one air-quality family and 72 episodes. Broader naturally incomplete data are needed before drawing deployment conclusions.

The primary mechanism tests use one registered router seed, although five fitted seeds yield identical core family-macro values. Candidate neural models use the repository’s common ten-epoch budget, and different budgets could alter the candidate ranking. TimesFM 2.5 and Sundial are run independently per target, whereas Chronos-2 receives multivariate context; this difference is part of the evaluated systems. The result does not isolate architecture from adapter behavior.

H4 and one required H6 component are unavailable. The negative joint decision follows the registered rule, while the missing comparisons reduce what the experiment can conclude about external selectors and fixed imputers. External-selector deltas lack confirmatory intervals because the development selection step stopped. LOFO evaluates unseen labelled families but still fits candidate imputers on the first 20% of each held-out series. This is unseen-family router transfer rather than strict zero-shot candidate transfer.

Finally, the study evaluates a frozen candidate registry, context and horizon of 96, a shortlist of six, and 20 forecast samples. It supplies empirical boundary evidence rather than a general impossibility result. Equivalence and the registered 1% non-inferiority margin were not tested as primary hypotheses.

9 CONCLUSION

We evaluated forecast supervision, block granularity, structured coordination, unseen-family transfer, and held-out-forecaster transfer as separate claims for B-FAIS. None of the six registered tests meets the superiority rule; four complete tests have intervals crossing zero, and two tests are incomplete under pre-specified availability conditions. Natural missingness also favors a complete fixed candidate over the current router for all three forecasters. The evidence narrows B-FAIS from a claimed structured selection advance to a tested system whose proposed mechanisms remain unvalidated. Further work should begin from the sequence-level control, predefine coverage-qualified comparator sets, and treat tail failures as primary outcomes.

AI USE STATEMENT

In preparing this revision, the authors used OpenAI Codex to inspect existing source code and machine-readable result artifacts; draft and edit LaTeX prose and tables; check citation keys, numerical consistency, and document formatting; and assist with interpretation of already generated experimental results. The tool did not generate datasets or the reported experimental measurements. All AI-assisted text and checks were reviewed against source code and saved artifacts by the authors. We take

responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The study uses publicly distributed numerical time-series datasets and adds synthetic masks to controlled confirmation data. It involves no intervention or human-subject recruitment. The Beijing data are distributed under CC BY 4.0; other dataset ownership and redistribution terms remain with their providers. Forecast errors can affect energy, labor, logistics, air-quality, and infrastructure decisions. The evaluated research system should not be used for autonomous operational decisions without domain-specific validation, monitoring, and fallback procedures.

REPRODUCIBILITY STATEMENT

Appendix A records data partitions, masks, seeds, candidates, and checkpoints. Appendix C gives component and sensitivity results, and Appendix D maps the reported claims to versioned artifacts and hash manifests. The supplementary source tree contains the frozen YAML configurations, evaluation code, tests, and machine-readable result summaries used for this paper.

REFERENCES

- Mauricio Morais Almeida, João Dallyson Sousa Almeida, Darlan Bruno Pontes Quintanilha, Geraldo Braz Júnior, and Aristófanés Correa Silva. A meta-learning based neural network and LSTM for univariate time series missing data imputation. *Applied Soft Computing*, 172:112845, 2025. doi: 10.1016/j.asoc.2025.112845. URL <https://doi.org/10.1016/j.asoc.2025.112845>.
- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=gerNCVqqtR>.
- Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. URL <https://arxiv.org/abs/2510.15821>.
- Christopher J. C. Burges. From RankNet to LambdaRank to LambdaMART: An overview. Technical Report MSR-TR-2010-82, Microsoft Research, 2010. URL <https://www.microsoft.com/en-us/research/publication/from-ranknet-to-lambdarank-to-lambdamart-an-overview/>.
- Wei Cao, Dong Wang, Jian Li, Hao Zhou, Lei Li, and Yitan Li. BRITS: Bidirectional recurrent imputation for time series. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://papers.nips.cc/paper_files/paper/2018/hash/734e6bfcd358e25ac1db0a4241b95651-Abstract.html.
- Song Chen. Beijing multi-site air quality. UCI Machine Learning Repository, 2017. URL <https://doi.org/10.24432/C5RK5G>.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 10148–10167. PMLR, 2024. URL <https://proceedings.mlr.press/v235/das24c.html>.

- Wenjie Du, David Côté, and Yan Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 219:119619, 2023. doi: 10.1016/j.eswa.2023.119619. URL <https://doi.org/10.1016/j.eswa.2023.119619>.
- Vincent Fortuin, Dmitry Baranchuk, Gunnar Rätsch, and Stephan Mandt. GP-VAE: Deep probabilistic time series imputation. In Silvia Chiappa and Roberto Calandra (eds.), *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pp. 1651–1661. PMLR, 2020. URL <https://proceedings.mlr.press/v108/fortuin20a.html>.
- Rakshitha W. Godahewa, Christoph Bergmeir, Geoffrey I. Webb, Rob J. Hyndman, and Pablo Montero-Manso. Monash time series forecasting archive. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/eddea82ad2755b24c4e168c5fc2ebd40-Paper-round2.pdf.
- Google Research. TimesFM. <https://github.com/google-research/timesfm>, 2025. TimesFM 2.5 software release documentation; accessed 28 August 2026.
- Hussein Hazimeh, Zhe Zhao, Aakanksha Chowdhery, Maheswaran Sathiamoorthy, Yihua Chen, Rahul Mazumder, Lichan Hong, and Ed H. Chi. DSelect-k: Differentiable selection in the mixture of experts with applications to multi-task learning. In *Advances in Neural Information Processing Systems*, volume 34, pp. 29335–29347. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/f5ac21cd0ef1b88e9848571aeb53551a-Paper.pdf.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. doi: 10.2307/4615733. URL <https://www.jstor.org/stable/4615733>.
- Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001. URL <https://doi.org/10.1016/j.ijforecast.2006.03.001>.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://papers.nips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html.
- Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, Caiyin Yang, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Sundial: A family of highly capable time series foundation models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 39295–39317. PMLR, 2025. URL <https://proceedings.mlr.press/v267/liu25be.html>.
- Rahul Mazumder, Trevor Hastie, and Robert Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *Journal of Machine Learning Research*, 11(80):2287–2322, 2010. URL <https://jmlr.org/papers/v11/mazumder10a.html>.
- Mustafa Mısırl and Michèle Sebag. ALORS: An algorithm recommender system. *Artificial Intelligence*, 244:291–314, 2017. doi: 10.1016/j.artint.2016.12.001. URL <https://doi.org/10.1016/j.artint.2016.12.001>.
- Tong Nie, Guoyang Qin, Wei Ma, Yuewen Mei, and Jian Sun. ImputeFormer: Low rankness-induced transformers for generalizable spatiotemporal imputation. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD ’24, pp. 2260–2271. Association for Computing Machinery, 2024. doi: 10.1145/3637528.3671751. URL <https://doi.org/10.1145/3637528.3671751>.
- John R. Rice. The algorithm selection problem. In *Advances in Computers*, volume 15, pp. 65–118. Elsevier, 1976. doi: 10.1016/S0065-2458(08)60520-3. URL [https://doi.org/10.1016/S0065-2458\(08\)60520-3](https://doi.org/10.1016/S0065-2458(08)60520-3).

- Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976. doi: 10.1093/biomet/63.3.581. URL <https://doi.org/10.1093/biomet/63.3.581>.
- Daniel J. Stekhoven and Peter Bühlmann. MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1):112–118, 2012. doi: 10.1093/bioinformatics/btr597. URL <https://doi.org/10.1093/bioinformatics/btr597>.
- Sabera J. Talukder, Yisong Yue, and Georgia Gkioxari. TOTEM: Tokenized time series EMbeddings for general time series analysis. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=Q1TLkH6xRC>.
- Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Advances in Neural Information Processing Systems*, volume 34, pp. 24804–24816. Curran Associates, Inc., 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/cfe8504bda37b575c70ee1a8276f3486-Abstract.html>.
- Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000. doi: 10.1016/S0169-2070(00)00065-0. URL [https://doi.org/10.1016/S0169-2070\(00\)00065-0](https://doi.org/10.1016/S0169-2070(00)00065-0).
- Stef van Buuren and Karin Groothuis-Oudshoorn. mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, 45(3):1–67, 2011. doi: 10.18637/jss.v045.i03. URL <https://doi.org/10.18637/jss.v045.i03>.
- Shiyu Wang, Jiawei Li, Xiaoming Shi, Zhou Ye, Baichuan Mo, Wenze Lin, Shengtong Ju, Zhixuan Chu, and Ming Jin. TimeMixer++: A general time series pattern machine for universal predictive analysis. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=1CLzLXSFNn>.
- Zhixian Wang, Linxiao Yang, Liang Sun, Qingsong Wen, and Yi Wang. Task-oriented time series imputation evaluation via generalized representers. In Amir Globerson, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 137403–137431. Curran Associates, Inc., 2024. doi: 10.52202/079017-4365. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/f88264fcc54775ee1706116e90fe351a-Abstract-Conference.html.
- Hsiang-Fu Yu, Nikhil Rao, and Inderjit S. Dhillon. Temporal regularized matrix factorization for high-dimensional time series prediction. In *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016. URL https://papers.nips.cc/paper_files/paper/2016/hash/85422afb467e9456013a2a51d4dff702-Abstract.html.
- Fengming Zhang, Wenjie Du, Huan Zhang, Ke Yu, and Shen Qu. HELIX: Hybrid encoding with learnable identity and cross-dimensional synthesis for time series imputation. In *Forty-Third International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=FN20iuPnEU>. Spotlight.
- Yue Zhao, Ryan Rossi, and Leman Akoglu. Automatic unsupervised outlier model selection. In *Advances in Neural Information Processing Systems*, volume 34, pp. 4489–4502. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/23c894276a2c5a16470e6a31f4618d73-Paper.pdf.
- Dongruo Zhou, Lihong Li, and Quanquan Gu. Neural contextual bandits with UCB-based exploration. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 11492–11502. PMLR, 2020. URL <https://proceedings.mlr.press/v119/zhou20a.html>.
- Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12):11106–11115, 2021. doi: 10.1609/aaai.v35i12.17325. URL <https://doi.org/10.1609/aaai.v35i12.17325>.

A DETAILED EXPERIMENTAL PROTOCOL

A.1 EPISODE CONSTRUCTION AND PARTITIONS

Table 4 records the fixed R2 episode protocol. Each complete source sequence is split temporally. Candidate imputers fit only on the first 20% prefix. Earlier eligible origins after that prefix generate router labels, and later origins form evaluation episodes. Synthetic masks are generated on the complete sequence before contexts are sliced, so methods share the same realization and an origin does not receive a favorable redraw.

Table 4: R2 episode construction. Dataset caps are reached in the common-16 and LOFO analyses.

Setting	Value
Context / horizon / forecast stride	96 / 96 / 96
Training-window stride	24
Candidate fit prefix	first 20% of each sequence
Forecast targets	registered variables 0 and 1
Items / train origins / evaluation origins	at most 4 / 4 / 4
Training episodes per dataset	at most 90
Evaluation episodes per dataset	90
Missing block lengths	6, 12, 24, 48
Missing rates	.1, .2, .3, .4, .5
Training mask seeds	1101, 1102, 1103
ETT development mask seeds	2101, 2102, 2103
Confirmation mask seeds	3101, 3102, 3103
Router seeds	4101, 4102, 4103, 4104, 4105
Global sampling seed	20260806
Forecast samples	20
Bootstrap replicates / seed	5,000 / 20260806

The six generators are random points, independent blocks, synchronous blocks, staggered correlated blocks, value-dependent blocks, and a mixture of independent blocks and random points. The labels describe deterministic stress-test implementations and do not assert the natural missing-data processes defined in statistical missingness theory (Rubin, 1976). Each dataset contributes one episode for every generator–rate–seed cell, yielding $6 \times 5 \times 3 = 90$ episodes. The common-16 population contains 2,340 episodes per forecaster. Its 140 zero-block windows result from slicing the registered full-sequence masks; every selector is a no-op on those windows. The remaining 2,200 windows contain at least one missing context cell.

A.2 DATASET INVENTORY

The ETT family (Zhou et al., 2021) is used only for development. Housing Inventory is excluded because 114 observations cannot support a 96-step context and 96-step future. Job Claims has one eligible evaluation origin. It cannot be separated from its rolling training origin, so common-16 excludes it; LOFO assigns that origin to held-out evaluation and includes a seventeenth family. Table 5 lists the 26 common-16 versions. Several sources are represented in the Monash forecasting archive (Godahewa et al., 2021); all source terms remain attached to the local data manifests.

A.3 CANDIDATE REGISTRY AND NATIVE VALIDITY

The registry contains 20 entries. TRMF is disabled because the installed PyPOTS 1.5 wrapper exposes a transductive interface that conflicts with rolling fit and inference. The other 19 candidates execute behind a common fit/impute interface. Candidate output is checked before the common numerical repair. A repaired value can keep a forecaster input finite, yet it never changes the candidate’s native-valid status.

The shortlist always contains LOCF and linear interpolation, then adds up to four candidates using predicted risk reduction divided by measured execution cost. Shortlisting is the only active compute penalty because the assignment cost coefficient is zero. Deep candidates use ten training epochs, batch size 16, and the repository-specific adapters. CSDI generates five imputation samples. These

Table 5: Common-16 confirmation families and dataset versions. Job Claims is added only to LOFO-17.

Family	Dataset versions
Azure 2019	azure2019_D_5T, azure2019_I_5T, azure2019_U_5T
Coastal TS	Coastal_T_S_15T, Coastal_T_S_20T, Coastal_T_S_H
Current velocity	current_velocity_5T, current_velocity_15T, current_velocity_20T, current_velocity_H
Electricity	electricity
EWELD load	EWELD_Load_15T
Exchange rate	exchange_rate
JOLTS	JOLTS_M
National illness	national_illness
NE China wind	NE_China_Wind_H
OpenElectricity NEM	OpenElectricity_NEM_5T
Port activity	Port_Activity_D, Port_Activity_W
Supply chain	Supply_Chain_Customer_D, Supply_Chain_Location_D
Traffic	traffic
Uncertainty	Uncertainty_1M_M
US labor	US_Labor_M
Vehicle	Vehicle_Sales_M, Vehicle_Supply_M
LOFO-only addition	Job_Claims_W

Table 6: Executed imputation candidates.

Group	Candidates
Temporal rules	LOCF, linear interpolation, seasonal lag
State-space and GP	Kalman local trend, Kalman AR, STL-Kalman, GP-RBF
Multivariate statistical	KNN, MICE, MissForest, SoftImpute
Specialized neural	BRITS, GP-VAE, SAITS, CSDI, ImputeFormer, HELIX
General time-series	TimeMixer++, TOTEM

common budgets make the experiment reproducible; they do not establish each candidate’s best attainable accuracy.

A.4 FORECASTER INTERFACES

The TimesFM adapter uses `google/timesfm-2.5-200m-pytorch` (Google Research, 2025) and predicts each target independently. Chronos-2 uses `amazon/chronos-2` (Ansari et al., 2025), receives a joint multivariate context, and supplies its median predictive quantile for point MASE. Sundial uses `thuml/sundial-base-128m` at snapshot `3212e42564493f520593e5414af4367fc4b49226` and predicts each target independently. Sundial supplies no teacher labels and is absent from ETT development.

B METHOD AND CONTROL DETAILS

B.1 TRAINING TARGETS

The teacher stores three forecast targets for a candidate row. Complete-candidate loss evaluates a context filled entirely by one candidate. Single-block loss replaces one block in a common anchor and records the resulting forecast loss. The coordination-adjusted target adds sampled interaction information. A separate reconstruction label measures masked-context absolute symmetric percentage error and is used only by SEQ-RECON and the external sequence selectors.

Table 7 gives the ETT development outcome that selected complete-candidate loss. These are development means, with no independent family replication. They justify configuration choice and are excluded from confirmatory claims. The learned-only evidence row uses the refined ranker score. The registered blend additionally uses a raw pseudo-error proxy and a historical candidate prior.

For the selected target, complete-candidate loss is repeated for candidate rows associated with sampled blocks. The ranker can condition on the block features, while the label itself evaluates the

Table 7: ETT development decisions. Lower MASE or negative learned-minus-blend difference is preferred. No confirmatory inference is attached to this table.

Ranker target	Chronos-2	TimesFM 2.5	Combined
Complete-candidate forecast loss	1.456	1.643	1.550
Single-block forecast loss	1.534	1.721	1.628
Coordination-adjusted target	1.763	1.706	1.735
Learned-only – registered blend	−.0348	−.0135	–

full candidate completion. This detail matters when interpreting H2: the block selector changes assignment granularity without introducing a locally observed block target.

B.2 FEATURE POLICIES

The deployment-available policy removes `dataset_id:*`, `family_id:*`, `missing_mechanism:*`, and the generator-provided target missing rate. Local and global missing fractions computed from the visible context remain available. The identity-free policy additionally removes `forecast_model:*`; candidate identity and candidate family remain because the selector must distinguish candidates. Table 8 lists the retained groups.

Table 8: Router feature groups. All quantities are available from training history, the corrupted context, candidate metadata, or candidate outputs.

Group	Examples
Block geometry	variable, start/end, length, relative position, boundary availability, local missing fraction
Episode context	visible missing fraction, variable count, seasonal period, training-prefix summaries
Candidate descriptors	candidate and family identity, fit scope, tail/period capability, stochastic flag, device and measured cost
Pseudo evidence	native coverage, MAE, RMSE, bias, uncertainty, output summaries, matched pseudo-block geometry
Pair descriptors	edge type and weight, overlap or gap, correlation, block-pair geometry, candidate pair, output summaries
Forecaster descriptors	forecaster identity, context mode, target granularity, output type; removed for identity-free transfer

B.3 GRAPH, PAIR LABELS, AND SOLVERS

Same-variable neighboring blocks receive a gap-decayed edge. Cross-variable blocks are connected when their time ranges overlap or touch, and nearby blocks can be connected when their variables are correlated in the fit prefix. Degree limits prevent dense graphs under synchronous point loss. Pair labels use a second-order counterfactual difference:

$$\begin{aligned}\Phi_{bc}(a, a') &= \mathcal{L}_f(A^{b \leftarrow a, c \leftarrow a'}, Y) - \mathcal{L}_f(A^{b \leftarrow a}, Y) \\ &\quad - \mathcal{L}_f(A^{c \leftarrow a'}, Y) + \mathcal{L}_f(A, Y),\end{aligned}\tag{5}$$

where A is a common complete anchor. At most four blocks, six candidates, and four pair labels are sampled per teacher episode. Structured inference uses beam width 32 and omits pair terms above 512 blocks. Independent block inference takes a valid per-block argmin. Sequence inference requires one candidate to be valid for every block and selects the candidate with lowest mean unary score.

B.4 EXTERNAL SELECTOR ADAPTATIONS

The external methods keep their sequence-level decision unit and reconstruction-based objective. MetaOD predicts a latent task representation; ALORS learns collaborative task–candidate factors; DSelect-1 uses one sparse expert gate; NeuralUCB uses a contextual-bandit score; random-valid

samples uniformly from candidates valid for the full episode. HybridLSTM uses fixed 48-step univariate windows and its method-specific pool. Appendix results report coverage separately because it lacks a valid output on some windows. Exact hyperparameters are frozen in `configs/iclr27-r2/baseline-selectors/`.

C FULL CONFIRMATORY AND SENSITIVITY RESULTS

C.1 SIX FIXED TESTS

Table 9 is the global multiplicity family. H1, H2, H3, and H5 contain every required component. H4 and H6 are incomplete under registered availability rules. Every adjusted value is one, and no row is eligible for a superiority statement.

Table 9: Global six-test decision table. An unavailable required component forces the joint raw value to one.

Test	Joint raw p	Holm p	Complete	Superiority eligible
H1: forecast versus reconstruction	.743561	1.000	yes	no
H2: block versus sequence	.463745	1.000	yes	no
H3: structured versus independent	.463745	1.000	yes	no
H4: core versus selected external	1.000000	1.000	no	no
H5: LOFO core versus sequence	.377823	1.000	yes	no
H6: Sundial transfer	1.000000	1.000	no	no

The H4 development selection covered all six registered external methods on ETT. HybridLSTM produced 100 ineligible primary episodes out of 360 for Chronos-2 and the same count for TimesFM 2.5. The registered rule `missing_duplicate_ineligible_or_nonfinite_primary_row_stops_decision` therefore stopped scoring and left the selected method null. For H6, the fixed-candidate branch found no candidate satisfying its availability floor. These outcomes are retained as unavailable components rather than replaced after observing confirmation results.

C.2 TAIL BEHAVIOR OF COMPLETE COMPONENTS

Table 10 reports paired episode degradation for every available fixed-test component. The median is zero for all rows because many candidate assignments coincide. Positive upper tails remain substantial for H1, H2, H5, and H6. H3 has the smallest tail differences, consistent with the near-identical structured and independent assignments.

Table 10: Tail statistics for tested-minus-reference MASE. Worse rate is the fraction with positive paired difference.

Test	Forecaster	Contrast	CVaR90	CVaR95	Maximum	Worse rate
H1	Chronos-2	Forecast — Recon	1.834	3.089	44.872	33.1%
H1	TimesFM 2.5	Forecast — Recon	1.586	2.710	44.219	30.5%
H2	Chronos-2	Block — Sequence	.901	1.555	29.616	34.4%
H2	TimesFM 2.5	Block — Sequence	.479	.798	5.009	24.4%
H3	Chronos-2	Structured — Independent	.133	.262	6.579	11.6%
H3	TimesFM 2.5	Structured — Independent	.024	.047	.927	4.2%
H5	Chronos-2	LOFO core — sequence	1.384	2.412	46.320	31.5%
H5	TimesFM 2.5	LOFO core — sequence	.684	1.085	5.088	27.2%
H6	Sundial	held-out core — sequence	.596	1.056	55.286	24.1%

C.3 DESCRIPTIVE EXTERNAL-SELECTOR COMPARISONS

Table 11 gives family-macro core-minus-external differences for pairable methods. The common-16 columns contain 2,340 paired episodes and 16 families; LOFO contains 2,430 episodes and 17 families. These rows have no confirmatory intervals because H4 selection stopped. HybridLSTM has 1,745 eligible common-16 episodes and 1,884 eligible LOFO episodes, which is below complete coverage.

Table 11: Descriptive family-macro differences against external sequence selectors. Negative values favor the B-FAIS core. C, T, and S denote Chronos-2, TimesFM 2.5, and Sundial.

Method	Common C	Common T	Common S	LOFO C	LOFO T
MetaOD	-.376	-.464	-.540	-.121	-.256
DSelect-1	-4.410	-3.615	-4.600	-1.998	-2.564
NeuralUCB	-.462	-.586	-.717	-.858	-.549
ALORS	-1.718	-1.236	-1.101	-.824	-.803
Random-valid	-2.320	-1.429	-1.986	-3.239	-2.586
HybridLSTM	unavailable	unavailable	unavailable	unavailable	unavailable

C.4 ROUTER-SEED VERIFICATION

All five rolling router seeds are present with distinct model and summary hashes. The resulting core family-macro MASE is identical at the reported precision: 7.516111 for Chronos-2 and 2.334864 for TimesFM 2.5 for seeds 4101 through 4105. The primary fixed tests use seed 4101 as registered. Seed identity is passed into both LambdaMART models and the pair regressor; it is not only encoded in the output directory name.

C.5 INCOMPLETE-PREFIX SENSITIVITY

The main synthetic protocol fits candidate models on a complete first-20% prefix. The sensitivity run corrupts that prefix at 10% and enforces that no training window is complete. Table 12 compares this run with the standard-prefix core under planned episode identities and distinct mask realization records. Family-macro intervals cross zero, while the upper episode tail grows sharply.

Table 12: No-complete-prefix minus standard-prefix sensitivity on 2,340 pairs and 16 families.

Forecaster	Family Δ	95% CI	CVaR95	Maximum	Worse rate
Chronos-2	+.078	$[-.055, .396]$	6.171	220.815	43.0%
TimesFM 2.5	+.018	$[-.075, .136]$	2.243	44.895	37.9%
Sundial	+.023	$[-.079, .141]$	1.878	39.890	37.8%

C.6 NATURAL-MISSING DETAILS

Every Beijing episode contains at least one native gap. The mean context missing fraction is 3.33%, with a range from 0.09% to 26.99%. Table 13 adds distributional statistics for B-FAIS. The oracle selects the lowest-MASE native-valid candidate per episode and is diagnostic only.

Table 13: Distribution of B-FAIS MASE on the 72 natural-missing episodes.

Forecaster	Mean	Median	P95	CVaR95	Oracle mean
Chronos-2	1.012	1.087	1.776	1.953	.943
TimesFM 2.5	1.127	1.152	2.114	2.498	1.018
Sundial	.909	.937	1.660	1.695	.786

D ARTIFACT INTEGRITY AND REPRODUCIBILITY

D.1 ARTIFACT MAP

The final integrity report has status VERIFIED. Table 14 maps manuscript evidence to repository-relative files. JSON and CSV versions are retained where both are available.

The H4 selection-stop file in Table 14 contains the stopping rule, ineligible counts, source hashes, and a null selected method. This file is required to interpret H4.

Table 14: Primary machine-readable evidence files.

Evidence	Repository-relative path
Integrity report	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/report.md
Six-test decisions	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/global_holm_tests.csv
Fixed-test components	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/global_holm_components.csv
External descriptions	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/external_selector_comparisons.csv
Evidence development	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/evidence_ablation_comparisons.csv
Prefix sensitivity	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/prefix_sensitivity_comparisons.csv
Natural missingness	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/uci_native_missing_summary.csv
Router-seed verification	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/common16_five_seed_verification.json
H4 selection stop	artifacts/iclr27-r2/r2-external-ett-development-selection-stopped-ms2101x3-rs4101-v005/selection.json
Resource summary	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/resource_summary.json
File and dependency hashes	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/source_and_dependency_signatures.json
Complete artifact manifest	artifacts/iclr27-r2/r2-final-analysis-integrity-v007/artifact_manifest.json

D.2 RESOURCE ACCOUNTING

The R2 tree contains 210 recorded runs and 84 evaluation manifests. All 84 evaluations are complete; 19 scheduler states record failed or interrupted attempts that are excluded from result summaries. Completed manifests record 145,639 forecaster calls over 2,669,387 forecast contexts. The maximum recorded CUDA allocation is 1,088,912,384 bytes and the maximum reservation is 1,166,016,512 bytes. Summed recorded wall time is 245.1 hours, including 81.8 hours of pipeline runtime and 9.6 hours inside forecast calls. These sums combine heterogeneous stages and should not be read as a single end-to-end benchmark.

D.3 LEGACY SEPARATION AND SOFTWARE CHECKS

R2 outputs are confined to `artifacts/iclr27-r2/`. The separation audit inventories 40,678 legacy files and finds zero legacy files written after the R2 start time. The legacy and R2 numbers are therefore not combined in any table in this paper.

The current source revision passes 526 automated tests. Static checks pass for Ruff and mypy over 61 source files. One expected MICE convergence warning is emitted by the test suite. These checks validate implemented invariants such as one-factor control configurations, mask partition separation, feature removal, native validity, routing metadata, and artifact signatures. They do not validate the scientific hypotheses, which are decided by the result tables above.

D.4 REPLICATION SCOPE

The source tree contains frozen configurations, registries, routing and evaluation code, result summaries, and tests. Exact reruns additionally require the local raw datasets, imputer checkpoints, and forecaster snapshots referenced by the manifests. No evaluation stage silently downloads a replacement forecaster. Observed context values are restored after assignment, and candidate eligibility is measured before fallback repair. A reproduction that changes either rule will change coverage and may change the paired population.

The recorded resource manifests identify Windows, Python 3.10, PyTorch 2.5.1 with CUDA 12.4, random seeds, resolved configurations, and input hashes. Raw dataset redistribution should follow each provider’s terms. The UCI Beijing dataset is CC BY 4.0 (Chen, 2017).